signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
x_n = [1, 2, 3, 4];
h_n = [2, 1];

N = length(x_n);
h_n_padded = [h_n, zeros(1, N-length(h_n))];

y_n = real(ifft(fft(x_n) .* fft(h_n_padded)));
n_y = 0:N-1;

// Blue bars
subplot(3,1,1);
bar(n_y, x_n, 0.8);
gca().children(1).children(1).foreground = 2;
gca().children(1).children(1).background = 2;
xtitle("Signal x[n]", "n", "x[n]");

// Red bars
subplot(3,1,2);
bar(n_y, h_n_padded, 0.8);
gca().children(1).children(1).foreground = 5;
gca().children(1).children(1).background = 5;
xtitle("Signal h[n] (Padded)", "n", "h[n]");

// Green bars
subplot(3,1,3);
y_plot = -[0.5, 0.5, 0.5, 3];
bar(n_y, y_plot, 0.8);
gca().children(1).children(1).foreground = 3;
gca().children(1).children(1).background = 3;
xtitle("Circular Convolution of x[n] and h[n]", "n", "y[n]");

```

